

A thermal-MPO derivation of the spin-1 gap bound

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From two certified finite-size purity estimates, we prove a uniform spectral gap $\Delta_L > (\log 20)/784$ for the periodic spin-1 antiferromagnetic Heisenberg chain for all even lengths $L \geq 2304$. The excited-state weight is bounded by the Gibbs-state purity, while an exact thermal matrix product operator (MPO) with a self-adjoint spatial transfer operator supplies a dual spatial purity. Combining operator squaring with spacetime coupling identities in an inductive doubling argument establishes the uniform lower bound.

Setting $J = k_B = 1$, the periodic spin-1 antiferromagnetic Heisenberg chain has Hamiltonian

$$H_L = \sum_{j=1}^L \mathbf{S}_j \cdot \mathbf{S}_{j+1}, \quad \mathbf{S}_{L+1} = \mathbf{S}_1, \quad (1)$$

for even $L \geq 4$, with ordered eigenvalues $E_0(L) \leq E_1(L) \leq \dots$ and spectral gap $\Delta_L = E_1(L) - E_0(L)$. A uniform spectral gap is a positive lower bound on Δ_L independent of L .

Theorem 1. *For every even $L \geq 2304$, the chain (1) has spectral gap $\Delta_L > (\log 20)/784 \simeq 0.003821$.*

The derivation combines two certified finite-size purities from Ref. [1] with three structural steps: the Gibbs purity bounding the gap [Eq. (4)], a continuous-time spatial transfer operator supplying a dual spatial purity [Eq. (6); Appendix], and an operator-squaring doubling iteration that drives both purities to one exponentially fast in β (Lemma 3, Proposition 1).

From purity to a gap.— At inverse temperature $\beta > 0$, the Gibbs density matrix and partition function are

$$\rho_L(\beta) = \frac{e^{-\beta H_L}}{Z_L(\beta)}, \quad Z_L(\beta) = \text{Tr } e^{-\beta H_L}. \quad (2)$$

With eigenvalues $w_m = e^{-\beta E_m(L)}/Z_L(\beta)$ ordered as $w_0 \geq w_1 \geq \dots$, the thermal purity

$$P_\tau(L, \beta) = \text{Tr } \rho_L(\beta)^2 = \frac{Z_L(2\beta)}{Z_L(\beta)^2} \quad (3)$$

satisfies $P_\tau = \sum_m w_m^2 \leq w_0 \sum_m w_m = w_0$, bounding the excited-state weight:

$$e^{-\beta \Delta_L} = \frac{w_1}{w_0} \leq \frac{1 - w_0}{w_0} \leq \frac{1 - P_\tau(L, \beta)}{P_\tau(L, \beta)}. \quad (4)$$

Establishing exponential decay of $1 - P_\tau(L, \beta)$ in β independent of L therefore yields a uniform gap.

Spatial transfer operator.— For each $\beta > 0$, the continuous-time thermal MPO (Appendix) yields a self-adjoint Hilbert–Schmidt operator X_β on $L^2(\mu_\beta)$ whose eigenvalues $\lambda_i(\beta)$ generate the partition functions for all even L :

$$Z_L(\beta) = \text{Tr}_v X_\beta^L = \sum_i |\lambda_i(\beta)|^L. \quad (5)$$

Tracing physical legs in the local tensor $W_\beta^{ss'}(a, b)$ gives the transfer vertex $X_\beta(a, b) = \sum_s W_\beta^{ss'}(a, b)$ and partition function $Z_n(\beta) = \text{Tr}_v X_\beta^n$. Normalizing the

open transfer product defines the spatial state $\sigma_n(\beta) = X_\beta^n/Z_n(\beta)$ and spatial purity

$$P_s(n, \beta) = \text{Tr}_v \sigma_n(\beta)^2 = \frac{Z_{2n}(\beta)}{Z_n(\beta)^2}. \quad (6)$$

For even n , self-adjointness ensures $X_\beta^n \geq 0$, so σ_n has unit trace and nonnegative spectrum, with $P_s(n, \beta) \leq 1$. While P_τ measures spectral concentration across temperatures, P_s reflects correlations across lengths.

Doubling iteration.— The doubling step combines an operator-squaring bound with an exact spacetime coupling identity. Squaring σ_n at fixed β doubles length, while squaring ρ_n doubles inverse temperature:

$$\sigma_n(\beta)^2 = P_s(n, \beta) \sigma_{2n}(\beta), \quad \rho_n(\beta)^2 = P_\tau(n, \beta) \rho_n(2\beta). \quad (7)$$

In both cases the spectrum is squared and renormalized: $p_i \mapsto p_i^2/P_s$ and $w_m \mapsto w_m^2/P_\tau$.

Lemma 1 (Squaring). *Let $p_i \geq 0$ with $\sum_i p_i = 1$ and purity $P = \sum_i p_i^2$. The squared spectrum p_i^2/P has purity P' satisfying*

$$1 - P' \leq f(1 - P), \quad f(z) = \frac{z^2}{2(1 - z)^2}. \quad (8)$$

Proof. Expanding $(\sum_i p_i)^2 = 1$ and $(\sum_i p_i^2)^2 = P^2$ gives $1 - P = 2 \sum_{i < k} p_i p_k$ and $1 - P' = 2 \sum_{i < k} p_i^2 p_k^2 / P^2$. Since all cross terms are nonnegative, $(1 - P)^2 = (2 \sum_{i < k} p_i p_k)^2 \geq 4 \sum_{i < k} p_i^2 p_k^2 = 2P^2(1 - P')$. $\square$

The function f is increasing on $[0, 1)$, with $f(z) \sim z^2/2$ as $z \rightarrow 0$, contracting the purity defect quadratically.

Lemma 2 (Coupling). *For all n and $\beta > 0$, the purities satisfy*

$$P_s(n, 2\beta) = \frac{P_s(n, \beta)^2 P_\tau(2n, \beta)}{P_\tau(n, \beta)^2}, \quad (9)$$

$$P_\tau(4n, \beta) = \frac{P_\tau(2n, \beta)^2 P_s(2n, 2\beta)}{P_s(2n, \beta)^2}. \quad (10)$$

Proof. Contracting four copies of $Z_n(\beta)$ into $Z_{2n}(2\beta)$ in space first gives $Z_n(\beta)^4 P_s(n, \beta)^2 P_\tau(2n, \beta)$, while in time first gives $Z_n(\beta)^4 P_\tau(n, \beta)^2 P_s(n, 2\beta)$:

$$\begin{array}{ccc} Z_n(\beta)^4 & \xrightarrow{\times P_s(n, \beta)^2} & Z_{2n}(\beta)^2 \\ \times P_\tau(n, \beta)^2 \downarrow & & \downarrow \times P_\tau(2n, \beta) \\ Z_n(2\beta)^2 & \xrightarrow{\times P_s(n, 2\beta)} & Z_{2n}(2\beta) \end{array}$$

Equating both paths gives Eq. (9); evaluating Eq. (9) at length $2n$ and solving for $P_\tau(4n, \beta)$ gives Eq. (10). $\square$

Lemma 3 (Doubling). *Assume Eq. (5), let n be even, and let $0 \leq \varepsilon \leq 1/100$. Then*

$$\begin{aligned} P_s(n, \beta), P_\tau(2n, \beta) &\geq 1 - \varepsilon \\ \implies P_s(2n, 2\beta), P_\tau(4n, 2\beta) &\geq 1 - 5\varepsilon^2. \end{aligned} \quad (11)$$

Proof. Spatial purity. Equation (9) with $P_\tau(n, \beta) \leq 1$ gives $P_s(n, 2\beta) \geq (1 - \varepsilon)^3$. Because n is even, $\sigma_n(2\beta)$ has nonnegative spectrum, and squaring it gives $\sigma_{2n}(2\beta)$. Lemma 1 and monotonicity of f therefore give $P_s(2n, 2\beta) \geq 1 - \delta$, where

$$\delta = f(1 - (1 - \varepsilon)^3) = \frac{1}{2}[(1 - \varepsilon)^{-3} - 1]^2 \leq 5\varepsilon^2 \leq \varepsilon, \quad (12)$$

using $1 - (1 - \varepsilon)^3 \leq 3\varepsilon$ and $9/[2(1 - 3\varepsilon)^2] \leq 5$ for $\varepsilon \leq 1/100$.

Thermal purity. Equation (10) with $P_s(2n, \beta) \leq 1$ and $\delta \leq \varepsilon$ gives $P_\tau(4n, \beta) \geq (1 - \varepsilon)^2(1 - \delta) \geq (1 - \varepsilon)^3$. Squaring $\rho_{4n}(\beta)$ gives $\rho_{4n}(2\beta)$, so Lemma 1 gives $P_\tau(4n, 2\beta) \geq 1 - \delta \geq 1 - 5\varepsilon^2$. $\square$

Induction and interpolation.—Iterating Lemma 3 from a single base scale yields a gap bound for all larger lengths.

Proposition 1 (Two-purity criterion [1]). *Assume Eq. (5). Let $n_0 \geq 4$ be even and $\beta_0 > 0$, with $P_s(n_0, \beta_0), P_\tau(2n_0, \beta_0) \geq 0.99$. Then $\Delta_L > (\log 20)/\beta_0$ for every even $L \geq n_0$.*

Proof. Induction on scale. For $j \geq 0$ set

$$n_j = n_0 2^j, \quad \beta_j = \beta_0 2^j, \quad \varepsilon_j = \frac{20^{-2^j}}{5}. \quad (13)$$

Induction on j establishes $P_s(n_j, \beta_j), P_\tau(2n_j, \beta_j) \geq 1 - \varepsilon_j$. The base case $j = 0$ is the hypothesis, since $\varepsilon_0 = 1/100$. Assuming the bound at j , Lemma 3 applies because $\varepsilon_j \leq 1/100$, yielding the bound at $j + 1$ since $2n_j = n_{j+1}$, $4n_j = 2n_{j+1}$, $2\beta_j = \beta_{j+1}$, and $5\varepsilon_j^2 = \varepsilon_{j+1}$. The defect decays exponentially in inverse temperature, $\varepsilon_j = \frac{1}{5}e^{-\beta_j(\log 20)/\beta_0}$, and doubly exponentially in the step count ($\varepsilon_4 \approx 3 \times 10^{-22}$).

Interpolation. Fix an even $L \geq n_0$ and choose j with $n_j \leq L \leq 2n_j$. Because ℓ^p norms decrease in p , the moment form (5) gives

$$Z_L(\beta_j) \leq Z_{n_j}(\beta_j)^{L/n_j}, \quad Z_L(2\beta_j) \geq Z_{2n_j}(2\beta_j)^{L/(2n_j)}, \quad (14)$$

for $L \geq n_j$ and $L \leq 2n_j$, respectively. Substituting into the thermal purity (3) gives

$$P_\tau(L, \beta_j) \geq [P_\tau(2n_j, \beta_j)P_s(n_j, \beta_j)^2]^{\frac{L}{2n_j}} \geq 1 - 3\varepsilon_j > \frac{1}{2}, \quad (15)$$

so Eq. (4) yields

$$e^{-\beta_j \Delta_L} \leq \frac{3\varepsilon_j}{1 - 3\varepsilon_j} \leq \frac{60}{97} 20^{-2^j} < 20^{-2^j}. \quad (16)$$

Taking logarithms and dividing by $\beta_j = \beta_0 2^j$ gives $\Delta_L > (\log 20)/\beta_0$. $\square$

Application to the Heisenberg chain.—

Proof of Theorem 1. By the Appendix, the Heisenberg chain satisfies the moment form (5). Proposition 5.5 of Ref. [1] certifies $P_s(2304, 784), P_\tau(4608, 784) \geq 0.99$; constant energy shifts cancel in purity ratios, so these bounds apply to H_L as written in Eq. (1). Proposition 1 with $(n_0, \beta_0) = (2304, 784)$ gives

$$\Delta_L > \frac{\log 20}{784} \simeq 0.003821 \quad (L \geq 2304 \text{ even}). \quad (17)$$

In particular, $\Delta_L > 0$, so the ground state is nondegenerate for every even $L \geq 2304$, and $\liminf \Delta_L \geq (\log 20)/784 > 0$ as $L \rightarrow \infty$ through even values. $\square$

Heisenberg transfer operator

The transfer representation (5) follows from the continuous-time MPO construction of Ref. [1] (Proposition 3.1). In the Cartesian basis $s, s' \in \{x, y, z\}$, the spin matrices $(S^\alpha)_{ss'} = -i\epsilon_{\alpha ss'}$ define real skew-symmetric operators $G_\alpha = iS^\alpha$ with $\|G_\alpha\| = 1$. The Hamiltonian is $-H_L = \sum_{j,\alpha} G_{\alpha,j} G_{\alpha,j+1}$, so each bond event (τ, α) in the exponential expansion inserts G_α at both endpoints. Letting a, b record bond events in imaginary time $0 < \tau < \beta$ on adjacent bonds, the continuous-time MPO tensor is

$$W_\beta^{ss'}(a, b) = \left\langle s \left| \mathcal{T}_\tau \prod_{(\tau, \alpha) \in a \cup b} G_\alpha \right| s' \right\rangle. \quad (\text{A.18})$$

Integrating over ordered imaginary times and summing event labels defines a finite measure μ_β of mass $e^{3\beta}$ (unit mass on the empty history). Contraction around the ring recovers the Dyson series as an exact MPO on $L^2(\mu_\beta)$ [2, 3].

Tracing physical legs yields the kernel $X_\beta(a, b) = \sum_s W_\beta^{ss}(a, b)$, which is real, symmetric, and bounded by 3. Since μ_β is finite, X_β is a self-adjoint Hilbert–Schmidt operator on $L^2(\mu_\beta)$. Closing the ring gives $Z_L(\beta) = \text{Tr}_v X_\beta^L$; real eigenvalues $\lambda_i(\beta)$ and even L ensure $\lambda_i(\beta)^L = |\lambda_i(\beta)|^L$, confirming Eq. (5).

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- [1] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Propositions 3.1, 4.3, and 5.5; Appendices A–C.
 - [2] W. Tang, H.-H. Tu, and L. Wang, [Continuous matrix product operator approach to finite temperature quantum states](#), Phys. Rev. Lett. **125**, 170604 (2020).
 - [3] B. Pirvu, V. Murg, J. I. Cirac, and F. Verstraete, [Matrix product operator representations](#), New J. Phys. **12**, 025012 (2010).